

Stabilizing Quantities ρ_p for Boundary Control of PDE

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1 Context

This is a question originally raised by Jean-Michel Coron in 2016. When studying the stability of nonlinear PDE of the form

$$\partial_t u + A(u)\partial_x u + B(u) = 0, \quad \text{on } [0, L], \quad (1)$$

where $u \in \mathbb{R}^n$ and with boundary conditions of the form

$$\begin{pmatrix} u^+(t, 0) \\ u^-(t, L) \end{pmatrix} = G \begin{pmatrix} u^+(t, L) \\ u^-(t, 0) \end{pmatrix}, \quad (2)$$

a natural quantity that appears is the following function

$$\rho_p : M \mapsto \inf \{ \|\Delta M \Delta^{-1}\|_p \mid \Delta = \text{diag}(\delta_1, \dots, \delta_n), \delta_i > 0, \forall i \in \{1, \dots, n\} \}, \quad (3)$$

where $\|\cdot\|_p$ is the matrix p -norm.

1.1 Conjectures

Conjecture 1. *For any $M \in \mathbb{R}^{n \times n}$ one has*

$$\rho_2(M) \leq \rho_3(M) \leq \dots \leq \rho_\infty(M). \quad (4)$$

A weaker version is:

Conjecture 2. *For any $M \in \mathbb{R}^{n \times n}$ one has*

$$\rho_2(M) \leq \rho_k(M), \quad \forall k \in \mathbb{N} \setminus \{0\}. \quad (5)$$

Another conjecture is:

Conjecture 3. *For any $n > 6$, and any $p > 2$ ($n, p \in \mathbb{N}$), there exists M such that*

$$\rho_p(M) < \rho_{p+1}(M). \quad (6)$$

2 What is Already Known

The main previous works are in [2, 1].

In particular it is known that for any $n \in \mathbb{N}$ and any $M \in \mathbb{R}^{n \times n}$,

$$\rho_1(M) \geq \rho_2(M), \quad \rho_1(M) = \rho_\infty(M), \quad (7)$$

and for $n \geq 2$ there exists $M \in \mathbb{R}^{n \times n}$ such that

$$\rho_1(M) > \rho_2(M). \quad (8)$$

It is also known [4] that if M has only non-negative coefficients then

$$\rho_k(M) = \rho_1(M), \quad \forall k \in [1, +\infty]. \quad (9)$$

3 Proofs

4 Preliminary Results

Throughout, $M \in \mathbb{R}^{n \times n}$ is fixed and $\|\cdot\|_p$ denotes the operator norm induced by the vector ℓ^p -norm on $\mathbb{R}^n$. We write $\mathcal{D}_{++}$ for the set of positive diagonal matrices.

4.1 Log-convexity

Lemma 1 (Interpolation of diagonal scalings). *Let $1 \leq p_0, p_1 \leq \infty$, $0 \leq \theta \leq 1$, and $\frac{1}{p_\theta} = \frac{1-\theta}{p_0} + \frac{\theta}{p_1}$. For any $\Delta_0, \Delta_1 \in \mathcal{D}_{++}$, define $\Delta_\theta := \Delta_0^{1-\theta} \Delta_1^\theta$ (entrywise). Then*

$$\|\Delta_\theta M \Delta_\theta^{-1}\|_{p_\theta} \leq \|\Delta_0 M \Delta_0^{-1}\|_{p_0}^{1-\theta} \|\Delta_1 M \Delta_1^{-1}\|_{p_1}^\theta.$$

Proof. Write $\Delta_j = D(h^{(j)})$ with

$$D(h) = \text{diag}(e^{h_1}, \dots, e^{h_n}), \quad h^{(j)} = (\log(\Delta_j)_{11}, \dots, \log(\Delta_j)_{nn}).$$

For z in the strip

$$S = \{z \in \mathbb{C} : 0 \leq \text{Re } z \leq 1\},$$

define

$$T_z = D((1-z)h^{(0)} + zh^{(1)}) M D(-((1-z)h^{(0)} + zh^{(1)})).$$

Each entry of T_z is of the form

$$(T_z)_{ij} = M_{ij} \exp((1-z)(h_i^{(0)} - h_j^{(0)}) + z(h_i^{(1)} - h_j^{(1)})),$$

hence is analytic in the open strip and continuous on the closed strip. Moreover,

$$|(T_z)_{ij}| \leq |M_{ij}| \exp(\max\{h_i^{(0)} - h_j^{(0)}, h_i^{(1)} - h_j^{(1)}\}), \quad z \in S,$$

so the family is uniformly bounded on S .

We now work temporarily on $\mathbb{C}^n$. On the boundary line $\text{Re } z = 0$, writing $z = iy$, we have

$$T_{iy} = U_y T_0 U_y^{-1}, \quad U_y = D(iy(h^{(1)} - h^{(0)})).$$

Since U_y is diagonal with unimodular entries, it is an isometry on every complex ℓ^p , and therefore

$$\|T_{iy}\|_{p_0, \mathbb{C}} = \|T_0\|_{p_0, \mathbb{C}}.$$

Similarly, on $\text{Re } z = 1$,

$$T_{1+iy} = V_y T_1 V_y^{-1}$$

with V_y diagonal unimodular, so

$$\|T_{1+iy}\|_{p_1, \mathbb{C}} = \|T_1\|_{p_1, \mathbb{C}}.$$

For a real matrix A , let $\|A\|_{p, \mathbb{R}}$ and $\|A\|_{p, \mathbb{C}}$ denote the induced operator norms on $\ell^p(\mathbb{R}^n)$ and $\ell^p(\mathbb{C}^n)$, respectively. Trivially,

$$\|A\|_{p, \mathbb{R}} \leq \|A\|_{p, \mathbb{C}}.$$

We claim that equality holds for every $1 \leq p \leq \infty$.

If $1 \leq p < \infty$, fix $z \in \mathbb{C}^n$ with $\|z\|_p = 1$, write $z_j = r_j e^{i\theta_j}$, and set

$$x_t = \text{Re}(e^{it} z) \in \mathbb{R}^n.$$

Because A is real,

$$(Ax_t)_i = \text{Re}(e^{it}(Az)_i).$$

Hence, with

$$c_p = \frac{1}{2\pi} \int_0^{2\pi} |\cos s|^p ds > 0,$$

we obtain using that $\cos$ is 2π -periodic and that $(Az)_j = |(Az)_j|e^{i\phi_j}$ for some $\phi_j \in [-\pi, \pi)$

$$\frac{1}{2\pi} \int_0^{2\pi} |(Ax_t)_i|^p dt = c_p |(Az)_i|^p, \quad \frac{1}{2\pi} \int_0^{2\pi} |(x_t)_j|^p dt = c_p r_j^p.$$

Summing over i and j gives

$$c_p \|Az\|_p^p = \frac{1}{2\pi} \int_0^{2\pi} \|Ax_t\|_p^p dt \leq \|A\|_{p,\mathbb{R}}^p \frac{1}{2\pi} \int_0^{2\pi} \|x_t\|_p^p dt = \|A\|_{p,\mathbb{R}}^p c_p.$$

Thus $\|Az\|_p \leq \|A\|_{p,\mathbb{R}}$, and taking the supremum over z yields

$$\|A\|_{p,\mathbb{C}} \leq \|A\|_{p,\mathbb{R}}.$$

If $p = \infty$, fix $z \in \mathbb{C}^n$ with $\|z\|_\infty = 1$. If $Az = 0$, there is nothing to prove. Otherwise choose i_0 such that

$$|(Az)_{i_0}| = \|Az\|_\infty.$$

Let $t = -\arg((Az)_{i_0})$ and set

$$x = \operatorname{Re}(e^{it}z) \in \mathbb{R}^n.$$

Then $\|x\|_\infty \leq 1$, and since A is real,

$$(Ax)_{i_0} = \operatorname{Re}(e^{it}(Az)_{i_0}),$$

so

$$\|Az\|_\infty = |(Az)_{i_0}| = |\operatorname{Re}(e^{it}(Az)_{i_0})| = |(Ax)_{i_0}| \leq \|Ax\|_\infty \leq \|A\|_{\infty,\mathbb{R}}.$$

Taking the supremum over z gives

$$\|A\|_{\infty,\mathbb{C}} \leq \|A\|_{\infty,\mathbb{R}}.$$

Therefore

$$\|A\|_{p,\mathbb{C}} = \|A\|_{p,\mathbb{R}} \quad \text{for every real matrix } A \text{ and every } 1 \leq p \leq \infty.$$

The family $\{T_z\}$ now satisfies the hypotheses of Stein's interpolation theorem for analytic families on the strip (see, e.g., [3, Ch. 4]). Applying it on the complex spaces $\ell^{p_0}(\mathbb{C}^n)$ and $\ell^{p_1}(\mathbb{C}^n)$ yields

$$\|T_\theta\|_{p_\theta,\mathbb{C}} \leq \|T_0\|_{p_0,\mathbb{C}}^{1-\theta} \|T_1\|_{p_1,\mathbb{C}}^\theta.$$

Since T_0 , T_1 , and T_θ are real matrices, the preceding equality of real and complex operator norms gives

$$\|T_\theta\|_{p_\theta} \leq \|T_0\|_{p_0}^{1-\theta} \|T_1\|_{p_1}^\theta.$$

Finally,

$$T_\theta = D((1-\theta)h^{(0)} + \theta h^{(1)}) M D(-(1-\theta)h^{(0)} + \theta h^{(1)}) = \Delta_\theta M \Delta_\theta^{-1},$$

so the lemma follows. $\square$

Proposition 2 (Log-convexity of ρ_p). *For any $M \in \mathbb{R}^{n \times n}$ and any $1 \leq p_0, p_1 \leq \infty$,*

$$\rho_{p_\theta}(M) \leq \rho_{p_0}(M)^{1-\theta} \rho_{p_1}(M)^\theta, \quad \frac{1}{p_\theta} = \frac{1-\theta}{p_0} + \frac{\theta}{p_1}.$$

Equivalently, with the convention $1/0 = \infty$ and $\log 0 = -\infty$, the map

$$t \mapsto \log \rho_{1/t}(M), \quad t \in [0, 1],$$

is convex on $[0, 1]$ as an extended-real-valued function.

Proof. Fix $\varepsilon > 0$. Choose $\Delta_0, \Delta_1 \in \mathcal{D}_{++}$ such that

$$\|\Delta_j M \Delta_j^{-1}\|_{p_j} \leq \rho_{p_j}(M) + \varepsilon, \quad j = 0, 1.$$

By Lemma 1,

$$\|\Delta_\theta M \Delta_\theta^{-1}\|_{p_\theta} \leq (\rho_{p_0}(M) + \varepsilon)^{1-\theta} (\rho_{p_1}(M) + \varepsilon)^\theta.$$

Since $\Delta_\theta \in \mathcal{D}_{++}$, the left-hand side is at least $\rho_{p_\theta}(M)$. Therefore

$$\rho_{p_\theta}(M) \leq (\rho_{p_0}(M) + \varepsilon)^{1-\theta} (\rho_{p_1}(M) + \varepsilon)^\theta.$$

Letting $\varepsilon \downarrow 0$ gives the stated inequality.

For the logarithmic reformulation, if $\rho_p(M) > 0$ for every p , then the preceding inequality is exactly the convexity of $t \mapsto \log \rho_{1/t}(M)$ with $t = 1/p$. If $\rho_{p_*}(M) = 0$ for some $p_* \in [1, \infty]$, then for every $1 \leq q \leq \infty$ and every matrix A one has

$$\|x\|_q \leq n^{\max\{0, 1/q-1/p_*\}} \|x\|_{p_*}, \quad \|x\|_{p_*} \leq n^{\max\{0, 1/p_*-1/q\}} \|x\|_q,$$

hence

$$\|Ax\|_q \leq n^{\max\{0, 1/q-1/p_*\}} \|Ax\|_{p_*} \leq n^{|1/p_*-1/q|} \|A\|_{p_*} \|x\|_q.$$

Therefore

$$\|A\|_q \leq n^{|1/p_*-1/q|} \|A\|_{p_*}.$$

Applying this to $A = \Delta M \Delta^{-1}$ and taking infima over $\Delta \in \mathcal{D}_{++}$ gives

$$\rho_q(M) \leq n^{|1/p_*-1/q|} \rho_{p_*}(M) = 0.$$

So $\rho_q(M) = 0$ for every q , and consequently $t \mapsto \log \rho_{1/t}(M)$ is identically $-\infty$, which is convex by convention. $\square$

4.2 Duality

Lemma 3 (Duality). *For $1/p + 1/q = 1$, $\rho_p(M) = \rho_q(M^T)$. In particular, $\rho_2(M) = \rho_2(M^T)$.*

Proof. By the norm duality $\|A\|_p = \|A^T\|_q$, we have

$$\|\Delta M \Delta^{-1}\|_p = \|(\Delta M \Delta^{-1})^T\|_q = \|\Delta^{-1} M^T \Delta\|_q$$

for any $\Delta \in \mathcal{D}_{++}$. Since $\Delta \mapsto \Delta^{-1}$ is a bijection of $\mathcal{D}_{++}$, taking infima gives $\rho_p(M) = \rho_q(M^T)$. $\square$

Corollary 4. *To prove $\rho_2(M) \leq \rho_p(M)$ for all M and all $p \geq 1$, it suffices to prove it for $p > 2$.*

Proof. The case $p = 1$ is already known: $\rho_1(M) \geq \rho_2(M)$ by (7). The case $p = 2$ is tautological. If $1 < p < 2$ and $q > 2$ is the conjugate exponent, then Lemma 3 gives

$$\rho_p(M) = \rho_q(M^T) \geq \rho_2(M^T) = \rho_2(M).$$

Thus only the range $p > 2$ remains. $\square$

4.3 Block-triangular reduction

We recall the standard *Frobenius normal form* (see, e.g., [5]): for any $M \in \mathbb{R}^{n \times n}$, there exists a permutation matrix P such that PMP^T is block upper triangular with irreducible diagonal blocks $M_{11}, \dots, M_{ss}$. Here “irreducible” means the directed support graph of the block is strongly connected (or the block is 1×1). This decomposition corresponds to the strongly connected components of the support graph of M .

Lemma 5 (Block reduction). *If, after permutation, M is block triangular (upper or lower) with diagonal blocks $M_{11}, \dots, M_{ss}$, then $\rho_p(M) = \max_\alpha \rho_p(M_{\alpha\alpha})$ for all $p \geq 1$.*

Proof. First treat the case where M is block upper triangular.

Lower bound. Every diagonal matrix is block diagonal with respect to the given partition, so write

$$\Delta = \text{diag}(\Delta_1, \dots, \Delta_s), \quad \Delta_\alpha \in \mathcal{D}_{++}.$$

Then

$$B := \Delta M \Delta^{-1}$$

is block upper triangular with diagonal blocks

$$B_{\alpha\alpha} = \Delta_\alpha M_{\alpha\alpha} \Delta_\alpha^{-1}.$$

Fix α and let $x = (0, \dots, 0, x_\alpha, 0, \dots, 0)$ be supported in block α . Then the α -th output block of Bx is exactly $B_{\alpha\alpha}x_\alpha$, so

$$\|Bx\|_p \geq \|B_{\alpha\alpha}x_\alpha\|_p.$$

Taking the supremum over all x_α with $\|x_\alpha\|_p = 1$ yields

$$\|B\|_p \geq \|B_{\alpha\alpha}\|_p = \|\Delta_\alpha M_{\alpha\alpha} \Delta_\alpha^{-1}\|_p.$$

Since this holds for every α and every Δ ,

$$\rho_p(M) = \inf_{\Delta \in \mathcal{D}_{++}} \|\Delta M \Delta^{-1}\|_p \geq \rho_p(M_{\alpha\alpha})$$

for each α . Therefore

$$\rho_p(M) \geq \max_{\alpha} \rho_p(M_{\alpha\alpha}).$$

Upper bound. Fix $\varepsilon > 0$ and choose $\Delta_\alpha \in \mathcal{D}_{++}$ such that

$$\|\Delta_\alpha M_{\alpha\alpha} \Delta_\alpha^{-1}\|_p \leq \rho_p(M_{\alpha\alpha}) + \varepsilon, \quad \alpha = 1, \dots, s.$$

Set

$$B_\alpha := \Delta_\alpha M_{\alpha\alpha} \Delta_\alpha^{-1}, \quad B_0 := \text{diag}(B_1, \dots, B_s),$$

and define, for $\eta > 0$,

$$\tilde{\Delta}(\eta) = \text{diag}(\Delta_1, \eta^{-1} \Delta_2, \dots, \eta^{-(s-1)} \Delta_s).$$

Then

$$B(\eta) := \tilde{\Delta}(\eta) M \tilde{\Delta}(\eta)^{-1}$$

has diagonal blocks B_α , while its (α, β) block for $\alpha < \beta$ is multiplied by the factor $\eta^{\beta-\alpha}$. Hence

$$B(\eta) \longrightarrow B_0 \quad \text{entrywise as } \eta \downarrow 0.$$

We now justify passage to the norm. First, $A \mapsto \|A\|_p$ is continuous in the matrix entries. Indeed, for any matrices A, B ,

$$|\|A\|_p - \|B\|_p| \leq \|A - B\|_p \leq n^{|1/p - 1/2|} \|A - B\|_F.$$

Thus $B(\eta) \rightarrow B_0$ in entries implies

$$\|B(\eta)\|_p \longrightarrow \|B_0\|_p.$$

Second, for a block-diagonal matrix $D = \text{diag}(C_1, \dots, C_s)$,

$$\|D\|_p = \max_{\alpha} \|C_\alpha\|_p.$$

For the lower bound, test D on vectors supported in a single block. For the upper bound, if $1 \leq p < \infty$ and $x = (x_1, \dots, x_s)$, then

$$\|Dx\|_p^p = \sum_{\alpha=1}^s \|C_\alpha x_\alpha\|_p^p \leq \left(\max_{\alpha} \|C_\alpha\|_p^p \right) \sum_{\alpha=1}^s \|x_\alpha\|_p^p = \left(\max_{\alpha} \|C_\alpha\|_p \right)^p \|x\|_p^p.$$

If $p = \infty$, then

$$\|Dx\|_\infty = \max_\alpha \|C_\alpha x_\alpha\|_\infty \leq \left(\max_\alpha \|C_\alpha\|_\infty \right) \max_\alpha \|x_\alpha\|_\infty = \left(\max_\alpha \|C_\alpha\|_\infty \right) \|x\|_\infty.$$

This proves the formula.

Applying it to B_0 gives

$$\|B_0\|_p = \max_\alpha \|B_\alpha\|_p \leq \max_\alpha (\rho_p(M_{\alpha\alpha}) + \varepsilon).$$

Since $\tilde{\Delta}(\eta) \in \mathcal{D}_{++}$ for every $\eta > 0$,

$$\rho_p(M) \leq \|B(\eta)\|_p \quad \text{for every } \eta > 0.$$

Letting $\eta \downarrow 0$ and using continuity,

$$\rho_p(M) \leq \|B_0\|_p \leq \max_\alpha (\rho_p(M_{\alpha\alpha}) + \varepsilon).$$

Finally let $\varepsilon \downarrow 0$ to obtain

$$\rho_p(M) \leq \max_\alpha \rho_p(M_{\alpha\alpha}).$$

This proves the result for block upper triangular matrices. For block lower triangular matrices, conjugate by the order-reversing permutation matrix P , which turns block lower triangular form into block upper triangular form. We have

$$\rho_p(PMP^T) = \rho_p(M),$$

because $\Delta \mapsto P\Delta P^T$ is a bijection of $\mathcal{D}_{++}$ and permutation matrices are isometries on every ℓ^p . Thus

$$\inf_{\Delta \in \mathcal{D}_{++}} \|\Delta(PMP^T)\Delta^{-1}\|_p = \inf_{\Delta' \in \mathcal{D}_{++}} \|P\Delta'M(\Delta')^{-1}P^T\|_p = \inf_{\Delta' \in \mathcal{D}_{++}} \|\Delta'M(\Delta')^{-1}\|_p.$$

So the same conclusion holds for lower triangular matrices as well. $\square$

4.4 Upper semicontinuity

Lemma 6 (Upper semicontinuity of ρ_p). *For any $p \geq 1$, $M \mapsto \rho_p(M)$ is upper semicontinuous: if $M_k \rightarrow M$ then*

$$\limsup_{k \rightarrow \infty} \rho_p(M_k) \leq \rho_p(M).$$

Proof. Fix $\Delta \in \mathcal{D}_{++}$. Since

$$\Delta M_k \Delta^{-1} \rightarrow \Delta M \Delta^{-1}$$

entrywise and the operator p -norm is continuous in the matrix entries,

$$\rho_p(M_k) \leq \|\Delta M_k \Delta^{-1}\|_p \longrightarrow \|\Delta M \Delta^{-1}\|_p.$$

Hence

$$\limsup_{k \rightarrow \infty} \rho_p(M_k) \leq \|\Delta M \Delta^{-1}\|_p.$$

Taking the infimum over $\Delta \in \mathcal{D}_{++}$ gives

$$\limsup_{k \rightarrow \infty} \rho_p(M_k) \leq \rho_p(M).$$

$\square$

4.5 Existence of optimal scaling

Lemma 7 (Existence). *Let $M \neq 0$ have irreducible (strongly connected) support and $p \in (1, \infty)$. Define*

$$\Phi_p(h) = \log \|D(h)MD(h)^{-1}\|_p, \quad h \in \mathbb{R}^n,$$

where $D(h) = \text{diag}(e^{h_i})$. Then:

1. Φ_p is convex on $\mathbb{R}^n$ and constant along $h + c\mathbf{1}$.
2. Φ_p is coercive on $H_0 := \{h : \sum h_i = 0\}$.
3. The infimum defining $\rho_p(M)$ is attained.

Proof. The zero matrix is excluded only because Φ_p would otherwise be undefined; if $M = 0$, then $\rho_p(M) = 0$ and every $\Delta \in \mathcal{D}_{++}$ is optimal.

If $n = 1$, then $H_0 = \{0\}$ and

$$D(h)MD(h)^{-1} = M \quad \text{for all } h \in \mathbb{R},$$

so all three assertions are immediate. Assume from now on that $n \geq 2$.

Convexity follows from Lemma 1 with $p_0 = p_1 = p$. Indeed, for $h^{(0)}, h^{(1)} \in \mathbb{R}^n$,

$$\|D((1-\theta)h^{(0)} + \theta h^{(1)})MD(-(1-\theta)h^{(0)} - \theta h^{(1)})\|_p \leq \|D(h^{(0)})MD(-h^{(0)})\|_p^{1-\theta} \|D(h^{(1)})MD(-h^{(1)})\|_p^\theta,$$

and taking logarithms gives

$$\Phi_p((1-\theta)h^{(0)} + \theta h^{(1)}) \leq (1-\theta)\Phi_p(h^{(0)}) + \theta\Phi_p(h^{(1)}).$$

Also,

$$D(h + c\mathbf{1}) = e^c D(h),$$

so

$$D(h + c\mathbf{1})MD(h + c\mathbf{1})^{-1} = D(h)MD(h)^{-1},$$

hence $\Phi_p(h + c\mathbf{1}) = \Phi_p(h)$.

For coercivity, since H_0 is finite-dimensional, it suffices to show that

$$\Phi_p(h) \rightarrow +\infty \quad \text{whenever } h \in H_0 \text{ and } \|h\|_\infty \rightarrow \infty.$$

Because $M \neq 0$, the finite set of nonzero entries of M is nonempty, so

$$\gamma := \min\{|M_{ij}| : M_{ij} \neq 0\} > 0.$$

For any matrix B and any i, j ,

$$\|B\|_p \geq |B_{ij}|,$$

since one may test B on e_j and then take the i -th coordinate. Therefore, for every (i, j) in the support of M ,

$$\|D(h)MD(h)^{-1}\|_p \geq e^{h_i - h_j} |M_{ij}|.$$

Now let $h \in H_0$. If $h = 0$, there is nothing to prove. Otherwise choose indices a, b such that

$$h_a = \max_i h_i, \quad h_b = \min_i h_i.$$

Then $a \neq b$, and since $\sum_i h_i = 0$, we have $h_a \geq 0 \geq h_b$. Hence

$$h_a - h_b = \max_i h_i - \min_i h_i \geq \|h\|_\infty.$$

Because the support graph of M is strongly connected, there exists a directed path

$$a = v_0 \rightarrow v_1 \rightarrow \cdots \rightarrow v_\ell = b$$

with distinct vertices, so $1 \leq \ell \leq n-1$ and

$$M_{v_r v_{r+1}} \neq 0 \quad (r = 0, \dots, \ell-1).$$

By telescoping,

$$h_a - h_b = \sum_{r=0}^{\ell-1} (h_{v_r} - h_{v_{r+1}}).$$

Therefore there exists some r such that

$$h_{v_r} - h_{v_{r+1}} \geq \frac{h_a - h_b}{\ell} \geq \frac{h_a - h_b}{n-1}.$$

For this edge,

$$\|D(h)MD(h)^{-1}\|_p \geq \gamma e^{h_{v_r} - h_{v_{r+1}}} \geq \gamma e^{(h_a - h_b)/(n-1)} \geq \gamma e^{\|h\|_\infty/(n-1)}.$$

Taking logarithms gives

$$\Phi_p(h) \geq \log \gamma + \frac{\|h\|_\infty}{n-1},$$

which tends to $+\infty$ as $\|h\|_\infty \rightarrow \infty$ on H_0 . Thus Φ_p is coercive on H_0 .

Finally, $h \mapsto D(h)MD(h)^{-1}$ is continuous in the entries, the operator norm is continuous, and $M \neq 0$, so Φ_p is continuous on $\mathbb{R}^n$. Since $\Phi_p(h + c\mathbf{1}) = \Phi_p(h)$, every $h \in \mathbb{R}^n$ has the same value as its unique translate

$$h - \frac{1}{n} \left(\sum_{i=1}^n h_i \right) \mathbf{1} \in H_0,$$

and therefore

$$\inf_{h \in \mathbb{R}^n} \Phi_p(h) = \inf_{h \in H_0} \Phi_p(h).$$

Consider the sublevel set

$$K := \{h \in H_0 : \Phi_p(h) \leq \Phi_p(0)\}.$$

By continuity, K is closed in H_0 ; by coercivity, K is bounded. Since H_0 is finite-dimensional, K is compact. By Weierstrass's theorem, the continuous function Φ_p attains its minimum on K , hence on H_0 , and therefore on $\mathbb{R}^n$.

If h_* is a minimizer and $\Delta_* = D(h_*)$, then

$$\|\Delta_* M \Delta_*^{-1}\|_p = e^{\Phi_p(h_*)} = \exp\left(\inf_{h \in \mathbb{R}^n} \Phi_p(h)\right) = \inf_{h \in \mathbb{R}^n} \|D(h)MD(h)^{-1}\|_p = \rho_p(M).$$

Thus the infimum defining $\rho_p(M)$ is attained. □

5 Proof of Conjecture 2: $\rho_2(M) \leq \rho_p(M)$

Theorem 8 (Conjecture 2). *For all $n \geq 1$, all $M \in \mathbb{R}^{n \times n}$, and all $p \geq 1$:*

$$\rho_2(M) \leq \rho_p(M).$$

The proof proceeds in three steps. First, we reduce the conjecture to a universal matrix inequality (Proposition 9). Then we prove this inequality using an SDP alternative theorem and the probabilistic method on the unit sphere.

5.1 Reduction to a universal inequality

Proposition 9. *For every $A \in \mathbb{R}^{n \times n}$ and every $p \in [1, \infty)$:*

$$\rho_2(A) \leq \|A\|_p.$$

Before proving Proposition 9, let us show it implies Theorem 8.

Proof that Proposition 9 implies Theorem 8. For $p \in [1, \infty)$: since ρ_2 is invariant under diagonal similarity (if Δ is a positive diagonal matrix, then $\rho_2(\Delta M \Delta^{-1}) = \rho_2(M)$, because $\{\Delta' \Delta M \Delta^{-1} \Delta'^{-1} : \Delta' \text{ pos. diag.}\} = \{\Delta'' M \Delta''^{-1} : \Delta'' \text{ pos. diag.}\}$), Proposition 9 gives

$$\rho_2(M) = \rho_2(\Delta M \Delta^{-1}) \leq \|\Delta M \Delta^{-1}\|_p$$

for every positive diagonal Δ . Taking the infimum over Δ yields $\rho_2(M) \leq \rho_p(M)$.

For $p = \infty$: the known result (7) gives $\rho_\infty(M) = \rho_1(M) \geq \rho_2(M)$. $\square$

5.2 SDP characterization of ρ_2 and its alternative

Lemma 10 (SDP characterization). *For $A \in \mathbb{R}^{n \times n}$:*

$$\rho_2(A) < 1 \iff \exists Q = \text{diag}(q_1, \dots, q_n) \succ 0 \text{ such that } A^T Q A \prec Q.$$

Proof. We have $\rho_2(A) = \inf_{\Delta} \|\Delta A \Delta^{-1}\|_2$ where Δ ranges over positive diagonal matrices. Setting $Q = \Delta^2$:

$$\|\Delta A \Delta^{-1}\|_2^2 = \lambda_{\max}((\Delta A \Delta^{-1})^T (\Delta A \Delta^{-1})) = \lambda_{\max}(\Delta^{-1} A^T \Delta^2 A \Delta^{-1}).$$

Hence $\|\Delta A \Delta^{-1}\|_2 < 1$ iff $\Delta^{-1} A^T \Delta^2 A \Delta^{-1} \prec I$, iff $A^T Q A \prec Q$ where $Q = \Delta^2$. Since $\Delta \mapsto Q = \Delta^2$ is a bijection of $\mathcal{D}_{++}$ onto itself (with inverse $Q \mapsto \Delta = Q^{1/2}$, the entrywise positive square root), the equivalence between $\inf_{\Delta} \|\Delta A \Delta^{-1}\|_2 < 1$ and $\exists Q \in \mathcal{D}_{++}$ with $A^T Q A \prec Q$ follows. $\square$

Lemma 11 (SDP alternative). *For symmetric matrices $B_1, \dots, B_n \in \mathbb{R}^{m \times m}$, exactly one of the following holds:*

(I) *There exist $q_1, \dots, q_n > 0$ such that $\sum_{i=1}^n q_i B_i \succ 0$.*

(II) *There exists $Y \in \mathbb{R}^{m \times m}$, $Y \succeq 0$, $Y \neq 0$, such that $\text{Tr}(Y B_i) \leq 0$ for all $i = 1, \dots, n$.*

Proof. (I) and (II) cannot both hold. If (I) holds with weights $q_i > 0$ and (II) holds with $Y \succeq 0$, $Y \neq 0$, then

$$0 < \text{Tr}\left(Y \sum_i q_i B_i\right) = \sum_i q_i \text{Tr}(Y B_i) \leq 0,$$

a contradiction (the first inequality uses $Y \succeq 0$, $Y \neq 0$, and $\sum q_i B_i \succ 0$).

Not-(I) implies (II). Suppose (I) fails. Define the convex cone

$$\mathcal{C} = \left\{ \sum_{i=1}^n q_i B_i : q_i \geq 0 \right\} \subset \mathcal{S}^m,$$

where $\mathcal{S}^m$ is the space of real symmetric $m \times m$ matrices equipped with the trace inner product $\langle X, Z \rangle = \text{Tr}(XZ)$. The cone $\mathcal{C}$ is convex (as a conic combination of fixed matrices). The open cone $\mathcal{S}_{++}^m$ of positive definite matrices is convex and open.

We claim $\mathcal{C} \cap \mathcal{S}_{++}^m = \emptyset$. If some $S = \sum q_i B_i \in \mathcal{S}_{++}^m$ with $q \geq 0$, then for any $\varepsilon > 0$, $S' = \sum (q_i + \varepsilon) B_i = S + \varepsilon \sum B_i$ lies in $\mathcal{S}_{++}^m$ for ε small enough (since $\mathcal{S}_{++}^m$ is open) and has all coefficients $q_i + \varepsilon > 0$, contradicting the failure of (I).

By the Hahn–Banach separation theorem (in the form: if K is convex and U is open convex with $K \cap U = \emptyset$ in a finite-dimensional space, there exists a separating hyperplane; no closedness of K is required), there exist a nonzero $Y \in \mathcal{S}^m$ and $\alpha \in \mathbb{R}$ such that

$$\text{Tr}(YX) \leq \alpha \quad \text{for all } X \in \mathcal{C}, \quad \text{Tr}(YZ) \geq \alpha \quad \text{for all } Z \in \mathcal{S}_{++}^m.$$

Since $0 \in \mathcal{C}$, the first condition gives $\alpha \geq 0$. Since both $\mathcal{C}$ and $\mathcal{S}_{++}^m$ are cones (stable under positive scalar multiplication), we can sharpen: for any $X \in \mathcal{C}$ and $t > 0$, $tX \in \mathcal{C}$, so $t\text{Tr}(YX) \leq \alpha$ for all $t > 0$, which forces $\text{Tr}(YX) \leq 0$. Similarly, for any $Z \in \mathcal{S}_{++}^m$ and $t > 0$, $tZ \in \mathcal{S}_{++}^m$, so $t\text{Tr}(YZ) \geq \alpha \geq 0$; if $\text{Tr}(YZ) < 0$ for some Z , taking $t \rightarrow \infty$ yields a contradiction. Hence $\text{Tr}(YZ) \geq 0$ for all $Z \in \mathcal{S}_{++}^m$.

By continuity, $\text{Tr}(YZ) \geq 0$ for all $Z \in \overline{\mathcal{S}_{++}^m} = \mathcal{S}_+^m$. In particular, taking $Z = vv^T \in \mathcal{S}_+^m$ for any $v \in \mathbb{R}^m$ gives $v^T Y v \geq 0$, so $Y \succeq 0$. Since $Y \neq 0$ (it is a nonzero separating functional), condition (II) holds: taking $X = B_i$ (which corresponds to $q = e_i \in \mathbb{R}_{\geq 0}^n$) gives $\text{Tr}(YB_i) \leq 0$. $\square$

5.3 Proof of Proposition 9

Proof of Proposition 9. We may assume $A \neq 0$ (otherwise $\rho_2(A) = 0$ and the result is trivial). It suffices to show:

$$\rho_2(A) \geq 1 \implies \|A\|_p \geq 1. \quad (10)$$

Indeed, once (10) is established, for any $\varepsilon > 0$, applying it to $A/(\|A\|_p + \varepsilon)$ (which has $\|A/(\|A\|_p + \varepsilon)\|_p < 1$, hence $\rho_2(A/(\|A\|_p + \varepsilon)) < 1$ by contrapositive of (10)) gives, by positive homogeneity $\rho_2(cA) = |c|\rho_2(A)$ (which follows directly from $\|\Delta(cA)\Delta^{-1}\|_2 = |c|\|\Delta A\Delta^{-1}\|_2$), that $\rho_2(A) < \|A\|_p + \varepsilon$. Letting $\varepsilon \rightarrow 0$ yields $\rho_2(A) \leq \|A\|_p$.

Step 1: Dual certificate from SDP alternative. Assume $\rho_2(A) \geq 1$. By Lemma 10, there is no positive diagonal matrix Q with $A^T Q A \prec Q$. Writing $Q = \text{diag}(q_1, \dots, q_n)$ and

$$B_i = e_i e_i^T - A^T e_i e_i^T A, \quad i = 1, \dots, n,$$

the condition $Q - A^T Q A \succ 0$ is exactly $\sum_i q_i B_i \succ 0$. Since this system has no solution with $q_i > 0$, Lemma 11 provides a *dual certificate*:

$$\exists Y \succeq 0, Y \neq 0, \quad \text{with} \quad \text{Tr}(YB_i) \leq 0 \quad \forall i.$$

Expanding:

$$\text{Tr}(YB_i) = \text{Tr}(Y e_i e_i^T) - \text{Tr}(Y A^T e_i e_i^T A) = Y_{ii} - (AY A^T)_{ii}.$$

So the dual certificate satisfies:

$$(AY A^T)_{ii} \geq Y_{ii} \quad \text{for all } i = 1, \dots, n. \quad (11)$$

Step 2: Gram factorization. Factor $Y = VV^T$ where $V \in \mathbb{R}^{n \times k}$ with $k = \text{rank}(Y) \geq 1$. Let $v_i \in \mathbb{R}^k$ denote the i -th row of V and $w_i \in \mathbb{R}^k$ denote the i -th row of $W := AV$. Then:

$$Y_{ii} = \|v_i\|_2^2, \quad (AY A^T)_{ii} = \|w_i\|_2^2.$$

Inequality (11) becomes:

$$\|w_i\|_2 \geq \|v_i\|_2 \quad \text{for all } i = 1, \dots, n. \quad (12)$$

Step 3: Spherical averaging (probabilistic method). Let u be uniformly distributed on the unit sphere $S^{k-1} \subset \mathbb{R}^k$. For any fixed $z \in \mathbb{R}^k$, the rotational invariance of the uniform measure on S^{k-1} gives

$$\mathbb{E}[|z^T u|^p] = c_{k,p} \|z\|_2^p, \quad (13)$$

where $c_{k,p} := \mathbb{E}[|u_1|^p] > 0$ depends only on k and p (not on z). Indeed, writing $z = \|z\|_2 \hat{z}$ with $\|\hat{z}\|_2 = 1$, the random variable $\hat{z}^T u$ has the same distribution as u_1 by rotational invariance.

For each i , applying (13) to $z = w_i$ and $z = v_i$, and using (12) together with the monotonicity of $t \mapsto t^{p/2}$ on $[0, \infty)$ (which gives $\|w_i\|_2^p \geq \|v_i\|_2^p$):

$$\mathbb{E}[|w_i^T u|^p] = c_{k,p} \|w_i\|_2^p \geq c_{k,p} \|v_i\|_2^p = \mathbb{E}[|v_i^T u|^p].$$

Summing over $i = 1, \dots, n$ and setting $x(u) = Vu$:

$$\mathbb{E}[\|Ax(u)\|_p^p] = \sum_{i=1}^n \mathbb{E}[|w_i^T u|^p] \geq \sum_{i=1}^n \mathbb{E}[|v_i^T u|^p] = \mathbb{E}[\|x(u)\|_p^p]. \quad (14)$$

Step 4: Extracting a witness. Define $f(u) = \|Ax(u)\|_p^p - \|x(u)\|_p^p$ for $u \in S^{k-1}$. By (14), $\mathbb{E}[f(u)] \geq 0$. Since $V \neq 0$ (as $Y \neq 0$) and $k = \text{rank}(Y) \geq 1$, $\ker(V)$ is a proper linear subspace of $\mathbb{R}^k$ of dimension $d < k$. Its intersection $\ker(V) \cap S^{k-1}$ is either empty (if $d = 0$) or a great subsphere of dimension $d-1 < k-1$, which has $(k-1)$ -dimensional surface measure zero on S^{k-1} (since a smooth submanifold of dimension $< k-1$ has zero $(k-1)$ -dimensional Hausdorff measure). For $u \in \ker(V)$, we have $x(u) = 0$ and $Ax(u) = 0$, so $f(u) = 0$. If $f(u) < 0$ for (almost) every $u \notin \ker(V)$, then $\mathbb{E}[f(u)] < 0$, contradicting $\mathbb{E}[f(u)] \geq 0$. Therefore, there exists $u_0 \in S^{k-1} \setminus \ker(V)$ with $f(u_0) \geq 0$. Set $x_0 = Vu_0 \neq 0$. Then $\|Ax_0\|_p^p \geq \|x_0\|_p^p$, so

$$\|A\|_p \geq \frac{\|Ax_0\|_p}{\|x_0\|_p} \geq 1.$$

This completes the proof of (10) and hence of Proposition 9. $\square$

Remark 12. *The proof avoids direct analysis of norming vectors, magnitude matching conditions, or Hessian inequalities—all of which were shown to encounter fundamental obstructions. Instead, the SDP alternative theorem converts the absence of a diagonal quadratic Lyapunov certificate into a dual object Y carrying row-wise ℓ^2 information. The probabilistic method then transfers this ℓ^2 information to ℓ^p via the rotational invariance of the uniform measure on the sphere, exploiting the fact that the moment identity (13) holds uniformly across all directions.*

6 Proof of Conjecture 1: Monotonicity of ρ_p

Theorem 13 (Conjecture 1). *For all $M \in \mathbb{R}^{n \times n}$,*

$$\rho_2(M) \leq \rho_3(M) \leq \rho_4(M) \leq \dots \leq \rho_\infty(M).$$

More generally, the function $p \mapsto \rho_p(M)$ is nondecreasing on $[2, \infty]$ and nonincreasing on $[1, 2]$.

Proof. We first treat the degenerate case $\rho_p(M) = 0$.

Recall the standard comparison of ℓ^r -norms on $\mathbb{R}^n$: for $1 \leq r \leq s \leq \infty$,

$$\|x\|_s \leq \|x\|_r \leq n^{1/r-1/s} \|x\|_s \quad (x \in \mathbb{R}^n),$$

with the convention $1/\infty = 0$. From this we obtain the following comparison of induced operator norms: for every $B \in \mathbb{R}^{n \times n}$ and every $p, q \in [1, \infty]$,

$$\|B\|_q \leq n^{1/q-1/p} \|B\|_p.$$

Indeed, if $1 \leq p \leq q \leq \infty$, then for every $x \neq 0$,

$$\frac{\|Bx\|_q}{\|x\|_q} \leq \frac{\|Bx\|_p}{\|x\|_q} \leq \|B\|_p \frac{\|x\|_p}{\|x\|_q} \leq n^{1/p-1/q} \|B\|_p.$$

Taking the supremum over $x \neq 0$ gives

$$\|B\|_q \leq n^{1/p-1/q} \|B\|_p.$$

If instead $1 \leq q \leq p \leq \infty$, then for every $x \neq 0$,

$$\frac{\|Bx\|_q}{\|x\|_q} \leq n^{1/q-1/p} \frac{\|Bx\|_p}{\|x\|_q} \leq n^{1/q-1/p} \|B\|_p \frac{\|x\|_p}{\|x\|_q} \leq n^{1/q-1/p} \|B\|_p,$$

and hence

$$\|B\|_q \leq n^{1/q-1/p} \|B\|_p.$$

Combining the two cases yields the stated bound

$$\|B\|_q \leq n^{|1/q-1/p|} \|B\|_p.$$

Now apply this to $B = \Delta M \Delta^{-1}$ and take the infimum over all positive diagonal Δ :

$$\rho_q(M) = \inf_{\Delta \text{ positive diagonal}} \|\Delta M \Delta^{-1}\|_q \leq n^{|1/q-1/p|} \inf_{\Delta \text{ positive diagonal}} \|\Delta M \Delta^{-1}\|_p = n^{|1/q-1/p|} \rho_p(M).$$

Therefore, if $\rho_p(M) = 0$ for some p , then $\rho_q(M) = 0$ for every $q \in [1, \infty]$. In that case the theorem is immediate. Henceforth assume

$$\rho_p(M) > 0 \quad \text{for all } p \in [1, \infty].$$

Since $\Delta = I$ is admissible, also $\rho_p(M) \leq \|M\|_p < \infty$, so all logarithms below are well defined.

Define

$$f(t) = \log \rho_{1/t}(M), \quad t \in [0, 1],$$

where $1/0 := \infty$, so that $f(0) = \log \rho_\infty(M)$.

Step 1: f is convex. This is Proposition 2.

Step 2: f attains its global minimum at $t = \frac{1}{2}$. Let $t \in (0, 1]$ and set $p = 1/t \in [1, \infty)$. By Theorem 8,

$$\rho_2(M) \leq \rho_p(M) = \rho_{1/t}(M).$$

Since $\log$ is increasing,

$$f\left(\frac{1}{2}\right) \leq f(t) \quad (0 < t \leq 1).$$

For $t = 0$, Theorem 8 at $p = 1$ gives $\rho_2(M) \leq \rho_1(M)$, and the known identity $\rho_1(M) = \rho_\infty(M)$ yields

$$\rho_2(M) \leq \rho_\infty(M),$$

hence

$$f\left(\frac{1}{2}\right) \leq f(0).$$

Therefore f attains its global minimum on $[0, 1]$ at $t = \frac{1}{2}$.

Step 3: Monotonicity of f on each side of $\frac{1}{2}$. We now prove the standard convexity fact directly.

If $0 \leq s < t \leq \frac{1}{2}$, set

$$\lambda = \frac{t-s}{\frac{1}{2}-s} \in (0, 1], \quad \text{so that} \quad t = (1-\lambda)s + \lambda \frac{1}{2}.$$

By convexity,

$$f(t) \leq (1-\lambda)f(s) + \lambda f\left(\frac{1}{2}\right).$$

Since $f(\frac{1}{2}) \leq f(s)$ by Step 2, we get

$$f(t) \leq (1-\lambda)f(s) + \lambda f(s) = f(s).$$

Thus f is nonincreasing on $[0, \frac{1}{2}]$.

If $\frac{1}{2} \leq s < t \leq 1$, set

$$\mu = \frac{s-\frac{1}{2}}{t-\frac{1}{2}} \in [0, 1), \quad \text{so that} \quad s = (1-\mu)\frac{1}{2} + \mu t.$$

Again by convexity,

$$f(s) \leq (1-\mu)f\left(\frac{1}{2}\right) + \mu f(t).$$

Since $f(\frac{1}{2}) \leq f(t)$ by Step 2, it follows that

$$f(s) \leq (1 - \mu)f(t) + \mu f(t) = f(t).$$

Hence f is nondecreasing on $[\frac{1}{2}, 1]$.

Step 4: Translate this back to p . If $2 \leq p_1 \leq p_2 \leq \infty$, let $t_i = 1/p_i$ (with $1/\infty = 0$). Then

$$0 \leq t_2 \leq t_1 \leq \frac{1}{2}.$$

Since f is nonincreasing on $[0, \frac{1}{2}]$,

$$f(t_1) \leq f(t_2).$$

Exponentiating gives

$$\rho_{p_1}(M) \leq \rho_{p_2}(M).$$

Thus $p \mapsto \rho_p(M)$ is nondecreasing on $[2, \infty]$. In particular,

$$\rho_2(M) \leq \rho_3(M) \leq \rho_4(M) \leq \cdots \leq \rho_\infty(M).$$

If $1 \leq p_1 \leq p_2 \leq 2$, then with $t_i = 1/p_i$ we have

$$\frac{1}{2} \leq t_2 \leq t_1 \leq 1.$$

Since f is nondecreasing on $[\frac{1}{2}, 1]$,

$$f(t_2) \leq f(t_1),$$

and therefore

$$\rho_{p_2}(M) \leq \rho_{p_1}(M).$$

So $p \mapsto \rho_p(M)$ is nonincreasing on $[1, 2]$.

Finally, using the known identity $\rho_1(M) = \rho_\infty(M)$, we conclude that $p \mapsto \rho_p(M)$ decreases on $[1, 2]$, increases on $[2, \infty]$, and has global minimum at $p = 2$. $\square$

References

- [1] J.-M. Coron and H.-M. Nguyen, *Dissipative boundary conditions for nonlinear 1-D hyperbolic systems: sharp conditions through an approach via time-delay systems*, SIAM J. Math. Anal., **47**(3):2220–2240, 2015.
- [2] J.-M. Coron, G. Bastin, and B. d’Andréa-Novel, *Dissipative boundary conditions for one-dimensional nonlinear hyperbolic systems*, SIAM Journal on Control and Optimization, **47**(3):1460–1498, 2008.
- [3] J. Bergh and J. Löfström, *Interpolation Spaces: An Introduction*, Grundlehren der mathematischen Wissenschaften, vol. 223, Springer-Verlag, 1976.
- [4] S. M. Rump, *Optimal scaling for p -norms and componentwise distance to singularity*, IMA Journal of Numerical Analysis, **23**(1):1–9, 2003.
- [5] A. Berman and R. J. Plemmons, *Nonnegative Matrices in the Mathematical Sciences*, Classics in Applied Mathematics, vol. 9, SIAM, 1994.